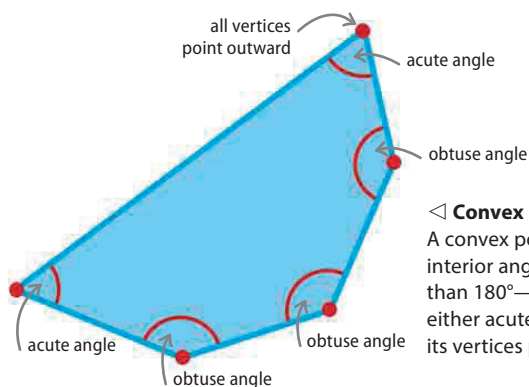


PROPERTIES OF A POLYGON

There are an unlimited number of different polygons that can be drawn using straight lines. However, they all share some important properties.

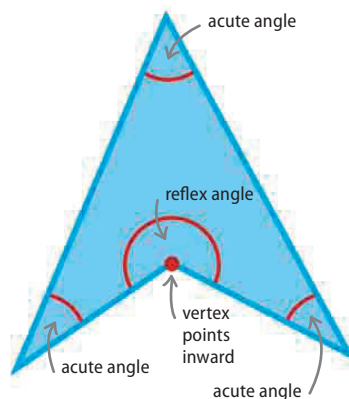
Convex or concave

Regardless of how many angles a polygon has, it can be classified as either concave or convex. This difference is based on whether a polygon's interior angles are over 180° or not. A convex polygon can be easily identified because at least one its angles is over 180° .



◁ Convex polygon

A convex polygon has no interior angles greater than 180° —its angles are either acute or obtuse. All its vertices point outward.

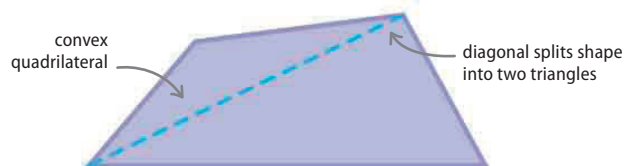


◁ Concave polygon

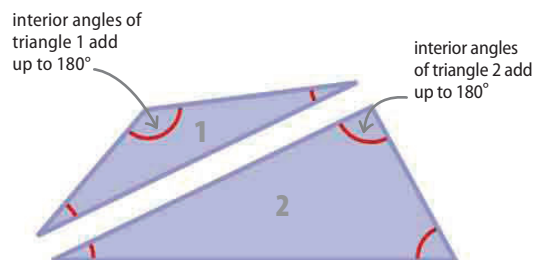
At least one angle of a concave polygon is over 180° . This type of angle is known as a reflex angle. The vertex of the reflex angle points inward, toward the center of the shape.

Interior angle sum of polygons

The sum of the interior angles of both regular and irregular convex polygon depends on the number of sides the polygon has. The sum of the angles can be worked out by dividing the polygon into triangles.

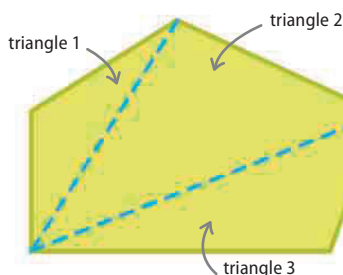


This quadrilateral is convex—all of its angles are smaller than 180° . The sum of its interior angles can be found easily, by breaking the shape down into triangles. This can be done by drawing in a diagonal line that connects two vertices that are not next to one another.



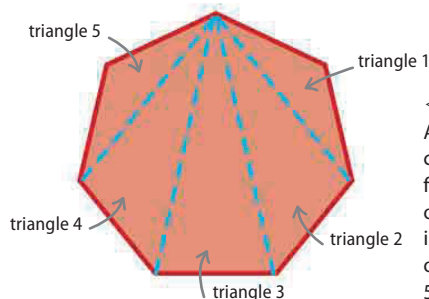
▶ A quadrilateral can be split into two triangles.

The sum of the angles of each triangle is 180° , so the sum of the angles of the quadrilateral is the sum of the angles of the two triangles added together: $2 \times 180^\circ = 360^\circ$.



◁ Irregular pentagon

This pentagon can be split up into three triangles. The sum of its interior angles is the sum of the angles of the three triangles: $3 \times 180^\circ = 540^\circ$.



◁ Regular heptagon

A heptagon (7 sides) can be split up into five triangles. The sum of its interior angles is the sum of the angles of the five triangles: $5 \times 180^\circ = 900^\circ$.